

Name: _____ Reg #: _____ Section: _____

National University of Computer and Emerging Sciences, Lahore Campus

Course: Artificial Intelligence

Program: BS(Computer Science)

Duration: 45 Minutes

Paper Date: 6-May-2026

Semester: Spring 2026

Course Code: AI2002

Section: BCS-6B

Total Marks: 20

Page(s): 2

Exam: Quiz 3

Q1) A smart factory has installed an AI system to detect defective products using sensor data. (3+2+3=8)

Input Features:

x1: Machine vibration intensity

x2: Temperature fluctuation level

For a product:

$$x_1 = 3, x_2 = 1$$

Hidden layer:

h₁: w₁₁ = 0.2, w₂₁ = -0.5, b₁ = 0.1

h₂: w₁₂ = -0.3, w₂₂ = 0.4, b₂ = -0.2

Activation: ReLU(z)

Output layer:

o: w₁ = 1.0, w₂ = -0.6, b = 0.1

Activation: ReLU

Target:

$$y = 1$$

Loss/Error: MSE

Hint (Derivatives): $\text{ReLU}'(z) = \{1 \text{ if } z > 0, 0 \text{ if } z \leq 0\}$

(a) Forward pass (compute all neuron outputs)

Unit	Net Input	Output
h ₁	$(0.2)(3) + (-0.5)(1) + 0.1 = 0.2$	$\max(0, 0.2) = 0.2$
h ₂	$(-0.3)(3) + (0.4)(1) - 0.2 = -0.7$	$\max(0, -0.7) = 0$
o	$(1.0)(0.2) + (-0.6)(0) + 0.1 = 0.3$	$\max(0, 0.3) = 0.3$

(b) Compute gradients for output weights w₁, w₂

	Gradients (T-O)(ReLU')(input)
$\partial E / \partial w_1$	$-(1-0.3)(1)(0.2) = -0.14$
$\partial E / \partial w_2$	$-(1-0.3)(0)(0) = 0$

(c) Update output weights and bias using learning rate $\eta = 0.1$

	New weights
w ₁	$1.0 - 0.1(-0.14) = 1.014$
w ₂	$-0.6 - 0.1(0) = -0.6$
b	$0.1 - 0.1(-0.7) = 0.17$

Q2) A tech company is developing a 3D object grouping system for an autonomous drone navigation module. Each detected object in space is represented using three spatial features: (2+2+2+2=8)

X: horizontal position, Y: vertical position & Z: depth (distance from drone)

The system uses K-Means clustering ($K = 2$) to group nearby objects into meaningful spatial regions for obstacle avoidance and path planning. The drone detects five objects in 3D space:

- $P1 \rightarrow (1, 2, 2)$
- $P2 \rightarrow (4, 4, 5)$
- $P3 \rightarrow (2, 3, 2)$
- $P4 \rightarrow (5, 5, 6)$
- $P5 \rightarrow (6, 5, 6)$

The clustering module initializes the system with the following grouping:

- Cluster C1: P1, P2, P3 (left-side spatial region)
- Cluster C2: P4, P5 (right-side spatial region)

(a) Compute the centroid (mean position) of each cluster to represent its “central navigation point” .

Centroid for Cluster 1	Centroid for Cluster 2
(2.33, 3, 3)	(5.5, 5, 6)

(b) The system identifies P2 as a boundary object between the two regions. Compute the Euclidean distance of P2 from both cluster centroids. Determine whether P2 should remain in C1 or be reassigned to C2 based on proximity.

Distance of P2 to Cluster 1	Distance of P2 to Cluster 2
2.79	2.06
Decision	P2 should be reassigned to Cluster C2

(c) If clusters are reassigned, update: calculate the updated centroids.

Centroid for Cluster 1	Centroid for Cluster 2
(1.5, 2.5, 2)	(5, 4.67, 5.67)

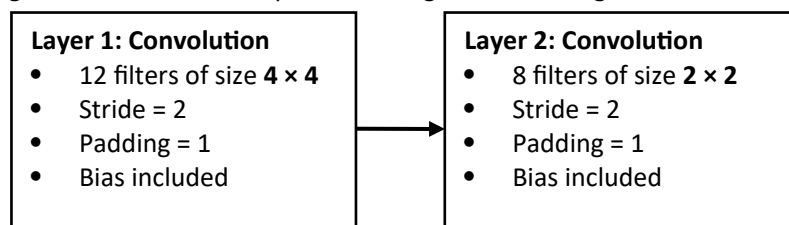
(d) During operation, the drone detects a sudden anomalous object $O \rightarrow (10, 1, 1)$ (possible sensor glitch or distant flying object). The system includes this object in clustering based on nearest centroid.

Which cluster will absorb this outlier?

Distance of O to Cluster 1	Distance of O to Cluster 2
8.69	7.76
Decision	Outlier joins Cluster C2

Q3) An RGB image of size $72 \times 72 \times 3$ is passed through the following CNN:

(4)



Calculate the following:

	Layer 1	Layer 2
Output Dimensions	$36 \times 36 \times 12$	$18 \times 18 \times 8$
Parameters	588	392